

Robust, Adaptive and Benchmark Attitude Control for a Low Earth Orbit Chaser Rendezvousing with the International Space Station

Technical Report: MATLAB ADCS Tool v1.1.0

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Problem Statement

This work implements and compares three 3-axis attitude controllers for a chaser spacecraft in a 400 km circular orbit that tracks a line-of-sight reference to the International Space Station (ISS). The controllers are a boundary-layer quaternion sliding-mode controller (robust, non-adaptive), a parametric adaptive sliding-mode controller that estimates the six inertia parameters online using a Slotine-Li regressor with box projection and a freeze-on-saturation anti-windup, and a quaternion proportional-derivative (PD) benchmark. The plant includes rigid-body dynamics with reaction-wheel gyroscopic coupling, torque- and momentum-limited wheels, gravity-gradient, aerodynamic, solar-radiation and magnetic disturbance torques, and optional sensor noise. The tool consists of a programmatically generated Simulink model, a MATLAB App Designer GUI and a MATLAB reference engine that runs the same library code. In MATLAB R2026a, the Simulink model and the reference engine agree to about 10^{-11}° on every deterministic scenario. Across seven scenarios and three controllers, both sliding-mode variants reduce the steady-state error roughly two-fold relative to the PD benchmark in the docking scenarios. In a persistently exciting attitude scan, the adaptive controller is about 75 times more accurate than PD and about 8 times more accurate than the robust controller.

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List of Abbreviations

ADCS	Attitude Determination and Control System
CG/CP	Center of Gravity / Center of Pressure
DCM	Direction Cosine Matrix
GUI	Graphical User Interface
ISS	International Space Station
LEO	Low Earth Orbit
LVLH	Local Vertical Local Horizontal
PD	Proportional-Derivative
PE	Persistent Excitation
RK4	Fourth-order Runge-Kutta integrator
SMC	Sliding Mode Control
SRP	Solar Radiation Pressure

1 Introduction

1.1 Problem and Scope

Autonomous rendezvous in low Earth orbit requires the chaser spacecraft to point its docking axis at the target while the actuators, the disturbance environment and the mass properties are all imperfectly known. This work compares three attitude control laws on a common platform so that their accuracy, actuator demand and robustness to inertia uncertainty can be evaluated under identical conditions. The chaser performs 3-axis rotational control only. The relative translational motion is represented by an analytic closing profile

$$R(t) = R_f + (R_0 - R_f) e^{-t/T}, \quad (1.1)$$

where $R_0 = 500$ m is the initial range, $R_f = 10$ m is the asymptotic final range and $T = 600$ s is the closing time constant. The radial and cross-track offsets shrink as $(R/R_0)^2$. This profile defines the docking-axis line of sight but is not integrated as coupled 6DOF dynamics. Four reference modes are available: inertial hold, LVLH hold, docking approach (default), and a 3-axis sinusoidal attitude scan about the LVLH hold that provides persistent excitation (PE) for the inertia estimator.

1.2 Related Work

Sliding-mode attitude control with a smooth boundary layer was introduced for spacecraft tracking maneuvers by Lo and Chen [1], and quaternion feedback for large-angle maneuvers has a long history in the PD form [2, 3]. Parametric adaptation of the inertia through the regressor formulation follows Slotine and Li [4] and Slotine and Di Benedetto [5]. Recent adaptive sliding-mode designs address event-triggered actuation [6], the unwinding problem [7] and flexible-body disturbances through observers [8]. This work does not propose

a new control law. It instead integrates established formulations into one verified comparison tool, and it characterizes how the adaptation gain and the excitation level interact in a rendezvous setting.

1.3 Contributions and Document Layout

The contributions of this work are:

1. A common spacecraft model and reference generator against which a robust SMC, an adaptive SMC and a PD benchmark are evaluated with identical actuator limits and disturbances.
2. An anti-windup mechanism for the inertia estimator that freezes the adaptation while the wheels are torque- or momentum-limited.
3. A Simulink model generated programmatically and verified against an independent MATLAB reference engine that runs the same library functions.
4. A set of seven scenarios that exercise nominal docking, inertia uncertainty, large-angle slews, disturbance stress, actuator saturation, sensor noise and persistent excitation.

Section 2 introduces the attitude kinematics, the plant dynamics, the reference modes and the disturbance models along with the baseline parameters. Section 3 derives the three control laws and the stability argument for the adaptive law. Section 4 describes the software implementation, and Section 5 documents the verification against the reference engine. Section 6 presents the scenario results, which are discussed in Section 7. Section 8 concludes with directions for future work.

2 Spacecraft Model

2.1 Attitude Kinematics

Quaternions are scalar-first, $\mathbf{q} = [q_0, \mathbf{q}_v^T]^T$, and describe the body frame B relative to the orbit-plane inertial frame N . The corresponding direction cosine matrix that transforms a vector from N to B is [9]

$$\mathbf{C}_{BN}(\mathbf{q}) = (q_0^2 - \mathbf{q}_v^T \mathbf{q}_v) \mathbf{I}_3 + 2 \mathbf{q}_v \mathbf{q}_v^T - 2 q_0 [\mathbf{q}_v \times], \quad (2.1)$$

where $[\mathbf{a} \times]$ is the skew-symmetric cross-product matrix of $\mathbf{a}$. The quaternion product is defined such that $\mathbf{C}(\mathbf{a} \otimes \mathbf{b}) = \mathbf{C}(\mathbf{a}) \mathbf{C}(\mathbf{b})$, and the conjugate is denoted $\mathbf{q}^*$. With the body rate $\boldsymbol{\omega}$ of B relative to N expressed in B axes, the DCM and the quaternion evolve as

$$\begin{aligned} \dot{\mathbf{C}}_{BN} &= -[\boldsymbol{\omega} \times] \mathbf{C}_{BN}, \\ \dot{\mathbf{q}} &= \frac{1}{2} \boldsymbol{\Xi}(\mathbf{q}) \boldsymbol{\omega} + k_n (1 - \mathbf{q}^T \mathbf{q}) \mathbf{q}, \\ \boldsymbol{\Xi}(\mathbf{q}) &= \begin{bmatrix} -\mathbf{q}_v^T \\ q_0 \mathbf{I}_3 + [\mathbf{q}_v \times] \end{bmatrix}, \end{aligned} \quad (2.2)$$

where $k_n = 1 \text{ s}^{-1}$ is a constraint-stabilization gain that drives the numerical quaternion norm back to unity. The normalization itself is applied in a separate block of the Simulink model to avoid an algebraic loop. Only the frames and quantities listed in Table 2.1 are used in the remainder of this work.

2.2 Rigid-Body Dynamics with Reaction Wheels

Three orthogonal reaction wheels are aligned with the body axes. The rotational dynamics of the spacecraft with the wheel gyroscopic coupling, and the wheel momentum balance, are

Table 2.1: Frames and principal symbols

Symbol	Definition	Unit
N, B, R	Orbit-plane inertial, body and reference (desired) frames	-
$\mathbf{q}, \mathbf{q}_r$	Attitude of B and of R relative to N	-
$\mathbf{q}_e$	Error quaternion, $\mathbf{q} \otimes \mathbf{q}_r^*$	-
$\boldsymbol{\omega}$	Body rate of B relative to N , in B axes	rad/s
$\boldsymbol{\omega}_r$	Reference rate of R relative to N , in R axes	rad/s
$\boldsymbol{\omega}_e$	Rate error, $\boldsymbol{\omega} - \mathbf{C}(\mathbf{q}_e)\boldsymbol{\omega}_r$	rad/s
$\mathbf{J}$	Spacecraft inertia matrix (true value)	kg m ²
$\mathbf{h}_w$	Reaction-wheel angular momentum, B axes	N m s
$\boldsymbol{\tau}_{rw}, \boldsymbol{\tau}_d$	Wheel torque on the body and disturbance torque	N m
$\boldsymbol{\theta}$	Inertia parameter vector $[J_{xx} \ J_{yy} \ J_{zz} \ J_{xy} \ J_{xz} \ J_{yz}]^T$	kg m ²
$\hat{\boldsymbol{\theta}}, \tilde{\boldsymbol{\theta}}$	Estimate of $\boldsymbol{\theta}$ and estimation error $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$	kg m ²
s	Sliding variable	rad/s
$\boldsymbol{\Lambda}, \mathbf{K}$	Sliding-surface slope and linear feedback gain	s ⁻¹ , N m s
$\boldsymbol{\eta}, \phi$	Switching gain and boundary-layer thickness	N m, rad/s
$\boldsymbol{\Gamma}$	Adaptation gain matrix	kg m ² s ²

given by

$$\begin{aligned} \mathbf{J} \dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega} + \mathbf{h}_w) + \boldsymbol{\tau}_{rw} + \boldsymbol{\tau}_d, \\ \dot{\mathbf{h}}_w &= -\boldsymbol{\tau}_{rw}, \end{aligned} \tag{2.3}$$

where $\mathbf{J}$ is the true inertia of the scenario. The wheel torque follows the commanded torque $\boldsymbol{\tau}_c$ through a per-axis saturation,

$$\boldsymbol{\tau}_{rw} = \text{sat}_{\tau_{\max}}(\boldsymbol{\tau}_c), \tag{2.4}$$

and the wheel momentum rate $\dot{h}_{w,i}$ is set to zero on any axis whose momentum has reached $\pm h_{\max}$ while the command would drive it further outward. In this work the limits are $\tau_{\max} = 0.15$ N m and $h_{\max} = 15$ N m s, which correspond to a derated HR12-25-class wheel [10].

2.3 Reference Modes and Error Variables

The reference generator provides $\mathbf{q}_r$, $\boldsymbol{\omega}_r$ and $\dot{\boldsymbol{\omega}}_r$ for the selected mode, where $\boldsymbol{\omega}_r$ is expressed in R axes. The attitude error is formed with the error quaternion,

$$\begin{aligned}\mathbf{q}_e &= \mathbf{q} \otimes \mathbf{q}_r^*, & \mathbf{C}(\mathbf{q}_e) &= \mathbf{C}_{BR}, \\ \boldsymbol{\omega}_e &= \boldsymbol{\omega} - \mathbf{C}(\mathbf{q}_e) \boldsymbol{\omega}_r,\end{aligned}\tag{2.5}$$

and the sign of $\mathbf{q}_e$ is flipped whenever $q_{e,0} < 0$ so that the shortest rotation is always selected. The scalar attitude error reported in the results is the principal rotation angle of $\mathbf{q}_e$, $e = 2 \operatorname{atan2}(\|\mathbf{q}_{e,v}\|, q_{e,0})$.

2.4 Disturbance Torques

Four environmental torques act on the spacecraft. The gravity-gradient torque for the orbital rate n and the nadir direction $\hat{\mathbf{r}}$ in the body frame is [11]

$$\boldsymbol{\tau}_{gg} = 3n^2 \hat{\mathbf{r}} \times (\mathbf{J}\hat{\mathbf{r}}).\tag{2.6}$$

The aerodynamic torque is the drag force with a center-of-pressure offset $\mathbf{r}_{cp}$ from the center of gravity, the solar radiation pressure (SRP) torque is computed analogously with the SRP center-of-pressure offset, and the magnetic torque is generated by a residual dipole $\mathbf{m}_{res}$ in an aligned-dipole geomagnetic field [12]:

$$\boldsymbol{\tau}_{aero} = \mathbf{r}_{cp} \times \mathbf{F}_{drag}, \quad \boldsymbol{\tau}_{srp} = \mathbf{r}_{cp,s} \times \mathbf{F}_{srp}, \quad \boldsymbol{\tau}_{mag} = \mathbf{m}_{res} \times \mathbf{B}.\tag{2.7}$$

The total disturbance is $\boldsymbol{\tau}_d = \boldsymbol{\tau}_{gg} + \boldsymbol{\tau}_{aero} + \boldsymbol{\tau}_{srp} + \boldsymbol{\tau}_{mag}$, where each term can be enabled and scaled independently. At a 45° misalignment the baseline model gives $\|\boldsymbol{\tau}_{gg}\| \approx 3.8 \times 10^{-3}$, $\|\boldsymbol{\tau}_{aero}\| \approx 1.9 \times 10^{-4}$, $\|\boldsymbol{\tau}_{srp}\| \approx 8 \times 10^{-6}$ and $\|\boldsymbol{\tau}_{mag}\| \approx 2 \times 10^{-5}$ N m. The gravity gradient

therefore dominates the environment by more than an order of magnitude.

2.5 Baseline Parameters

Table 2.2 lists the baseline spacecraft, orbit and environment parameters used by every scenario unless a scenario overrides them. The default inertia matrix is

$$\mathbf{J}_{nom} = \begin{bmatrix} 1200 & 100 & -200 \\ 100 & 2200 & 300 \\ -200 & 300 & 3100 \end{bmatrix} \text{ kg m}^2. \quad (2.8)$$

The atmospheric density at 400 km is set to 3×10^{-12} kg/m³, which is slightly above the U.S. Standard Atmosphere 1976 value of 2.8×10^{-12} kg/m³ [13] and well within the range of about 5×10^{-13} to 2×10^{-11} kg/m³ encountered over the solar cycle [14]. The SRP pressure is derived from a total solar irradiance of 1361 W/m² [15], and the dipole field strength is computed from the IGRF-14 coefficients [16].

3 Controller Design

3.1 Sliding Variable and Inertia Regressor

All three controllers use the error quaternion and the rate error defined in Equation 2.5, and all of them add the wheel gyroscopic compensation $+\boldsymbol{\omega} \times \mathbf{h}_w$ to the commanded torque so that the wheel coupling in Equation 2.3 is cancelled. The two SMC variants share the sliding variable and the Slotine-Li regressor [4, 5]. The sliding variable is formed from the rate error and the vector part of the error quaternion,

$$\begin{aligned} \mathbf{s} &= \boldsymbol{\omega}_e + \boldsymbol{\Lambda} \mathbf{q}_v^{(e)} = \boldsymbol{\omega} - \boldsymbol{\omega}_{rv}, \\ \boldsymbol{\omega}_{rv} &= \mathbf{C}(\mathbf{q}_e) \boldsymbol{\omega}_r - \boldsymbol{\Lambda} \mathbf{q}_v^{(e)}, \end{aligned} \quad (3.1)$$

Table 2.2: Baseline spacecraft, orbit and environment parameters

Parameter	Value	Unit
Orbit altitude	400	km
Orbit inclination	51.64	deg
Initial range R_0 , final range R_f	500, 10	m
Closing time constant T	600	s
Initial radial offset	100	m
Wheel torque limit $\tau_{\max}$	0.15	N m
Wheel momentum limit $h_{\max}$	15	N m s
Atmospheric density	3×10^{-12}	kg/m ³
Drag coefficient, drag area	2.2, 10	-, m ²
Aerodynamic CP offset (body axes)	[0.10, 0.05, 0]	m
SRP pressure at 1 AU	4.54×10^{-6}	N/m ²
Reflectance factor, SRP area	0.6, 10	-, m ²
SRP CP offset (body axes)	[0.05, 0.10, 0]	m
Equatorial dipole field strength B_0	2.97×10^{-5}	T
Residual magnetic dipole	[0.5, 0.5, 0.5]	A m ²
Integration step (RK4 / ode4)	0.1	s
Constraint-stabilization gain k_n	1.0	s ⁻¹
Star-tracker noise (when enabled)	0.005	deg (1 σ)
Gyro white noise (when enabled)	0.001	deg/s (1 σ)
Gyro bias random walk (when enabled)	10^{-5}	deg/s/ $\sqrt{s}$

where $\mathbf{q}_v^{(e)}$ is the vector part of $\mathbf{q}_e$ and $\mathbf{\Lambda}$ is a diagonal positive-definite matrix. The inertia is parametrized linearly through the vector $\boldsymbol{\theta}$ of its six independent entries, which gives the regressor relation

$$\mathbf{Y}(\boldsymbol{\omega}, \boldsymbol{\omega}_{rv}, \dot{\boldsymbol{\omega}}_{rv}) \boldsymbol{\theta} = \mathbf{J} \dot{\boldsymbol{\omega}}_{rv} + \boldsymbol{\omega}_{rv} \times (\mathbf{J} \boldsymbol{\omega}), \quad \mathbf{Y} \in \mathbb{R}^{3 \times 6}. \quad (3.2)$$

Substituting Equation 3.1 into Equation 2.3 and applying the gyroscopic compensation yields the sliding dynamics

$$\mathbf{J} \dot{\mathbf{s}} = (\mathbf{J} \boldsymbol{\omega}) \times \mathbf{s} + \mathbf{Y} \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \right) - \mathbf{K} \mathbf{s} - \boldsymbol{\eta} \circ \text{sat}(\boldsymbol{\phi}^{-1} \circ \mathbf{s}) + \boldsymbol{\tau}_d, \quad (3.3)$$

which is valid for the control laws of Section 3.2 and Section 3.3. Here $\circ$ denotes the element-wise product, and the nominal estimate $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_{nom}$ is used by the robust controller.

3.2 Robust Sliding-Mode Controller

The robust controller feeds forward the nominal inertia through the regressor and adds a linear term and a boundary-layer switching term on the sliding variable,

$$\boldsymbol{\tau}_c = \mathbf{Y}\boldsymbol{\theta}_{nom} - \mathbf{K}\mathbf{s} - \boldsymbol{\eta} \circ \text{sat}(\boldsymbol{\phi}^{-1} \circ \mathbf{s}) + \boldsymbol{\omega} \times \mathbf{h}_w. \quad (3.4)$$

The saturation function replaces the sign function to avoid chattering, and the boundary-layer thickness $\boldsymbol{\phi}$ trades tracking accuracy against control smoothness. The switching gain must satisfy $\eta_i \geq \sup|d_i|$ for the lumped disturbance $\mathbf{d}$, which includes the unmodelled inertia error and the environmental torques.

3.3 Adaptive Sliding-Mode Controller

The adaptive controller replaces the nominal inertia by an online estimate and updates the estimate with a gradient law,

$$\begin{aligned} \boldsymbol{\tau}_c &= \mathbf{Y}\hat{\boldsymbol{\theta}} - \mathbf{K}\mathbf{s} - \boldsymbol{\eta} \circ \text{sat}(\boldsymbol{\phi}^{-1} \circ \mathbf{s}) + \boldsymbol{\omega} \times \mathbf{h}_w, \\ \dot{\hat{\boldsymbol{\theta}}} &= \text{Proj}(-\boldsymbol{\Gamma}\mathbf{Y}^T\mathbf{s}), \end{aligned} \quad (3.5)$$

where $\boldsymbol{\Gamma}$ is a diagonal positive-definite adaptation gain. The operator $\text{Proj}(\cdot)$ keeps each component of $\hat{\boldsymbol{\theta}}$ inside a box $[\theta_{i,\min}, \theta_{i,\max}]$ around the nominal value by zeroing the offending component of the update at an active bound. The principal moments are bounded within a fraction ρ of their nominal values, and each product of inertia is bounded within $\pm(|\theta_{nom,i}| + \rho\bar{J})$ of the nominal product, where $\bar{J}$ is the mean principal moment. The default width is $\rho = 0.6$.

Anti-windup

The adaptation law of Equation 3.5 assumes that the commanded torque is applied to the plant. When the wheels are torque- or momentum-limited, the closed loop no longer matches the model behind the adaptive law, and the estimator integrates $\hat{\boldsymbol{\theta}}$ onto its projection bounds. In the first verification pass, this mechanism caused the adaptive controller to lose to the robust controller in both slew scenarios, with the estimate reaching 78 to 80% error after the wheels had been limited for 500 to 650 s. To prevent this, the update $\dot{\hat{\boldsymbol{\theta}}}$ is frozen whenever a torque command exceeds $\tau_{\max}$ or a wheel momentum reaches $h_{\max}$, in the spirit of the input-constrained adaptive designs of [17, 18]. This restores parity between the two controllers, as shown in Figure 6.4.

Stability argument

Consider the Lyapunov candidate

$$V = \frac{1}{2} \mathbf{s}^T \mathbf{J} \mathbf{s} + \frac{1}{2} \tilde{\boldsymbol{\theta}}^T \boldsymbol{\Gamma}^{-1} \tilde{\boldsymbol{\theta}}. \quad (3.6)$$

Because $\mathbf{s}^T ((\mathbf{J}\boldsymbol{\omega}) \times \mathbf{s}) = 0$ and $\boldsymbol{\theta}$ is constant, the derivative of V along Equation 3.3 with the update law of Equation 3.5 becomes

$$\dot{V} = -\mathbf{s}^T \mathbf{K} \mathbf{s} - \sum_{i=1}^3 \left[\eta_i s_i \text{sat}\left(\frac{s_i}{\phi_i}\right) - s_i d_i \right] \leq -\mathbf{s}^T \mathbf{K} \mathbf{s} \quad \text{for } |s_i| \geq \phi_i, \eta_i \geq |d_i|, \quad (3.7)$$

since the term $\tilde{\boldsymbol{\theta}}^T \mathbf{Y}^T \mathbf{s}$ is cancelled by the adaptation law and the projection never increases V when the true parameters lie inside the box. This yields the usual bounded-tracking result: $\mathbf{s}$ converges to the boundary layer and the tracking error is ultimately bounded. The controller of Equation 3.4 is written in regressor form so that the adaptive controller differs from it only by the replacement $\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}_{nom}$, and this identity is verified numerically by a unit test to 10^{-12} .

3.4 Quaternion PD Benchmark

The benchmark is the classical quaternion feedback law [2, 3] with a diagonal gain structure,

$$\boldsymbol{\tau}_c = -\mathbf{K}_p \mathbf{q}_v^{(e)} - \mathbf{K}_d \boldsymbol{\omega}_e + \boldsymbol{\omega} \times \mathbf{h}_w, \quad (3.8)$$

where the gains are derived from the desired bandwidth ω_n and damping ratio ζ as $K_{p,i} = 2J_{ii}\omega_n^2$ and $K_{d,i} = 2\zeta\omega_n J_{ii}$. The benchmark has no feedforward of the reference acceleration or of the inertia, which makes it a useful lower bound on the achievable accuracy.

3.5 Controller Gains

Table 3.1 lists the default gains. The gains of the robust and adaptive controllers are identical except for the adaptation gain, so that any difference between them is attributable to the adaptation. The adaptation gain is deliberately conservative in the non-PE scenarios, and the PE scenario raises it explicitly as discussed in Section 7.

Table 3.1: Default controller gains

Gain	Value	Unit
$\boldsymbol{\Lambda}$ (slope)	$0.05 \mathbf{I}_3$	s^{-1}
$\mathbf{K}$ (feedback)	$\text{diag}(\mathbf{J}_{nom})/20$	N m s
$\boldsymbol{\eta}$ (switching)	2×10^{-3} per axis	N m
$\boldsymbol{\phi}$ (boundary layer)	10^{-3} per axis	rad/s
$\boldsymbol{\Gamma}$ (adaptation)	2×10^7 per parameter	$\text{kg m}^2 \text{s}^2$
$\boldsymbol{\Gamma}$ in the PE scenario	5×10^9 per parameter	$\text{kg m}^2 \text{s}^2$
Projection width ρ	0.6	-
PD bandwidth ω_n	0.03	rad/s
PD damping ratio ζ	0.9	-

4 Software Implementation

The tool consists of a Simulink model, a MATLAB reference engine, a library of functions shared by both engines, and an App Designer GUI. Figure 4.1 shows the topology of the simulation harness. Every block calls the same library function in Simulink, through MATLAB Function blocks, and in the reference engine, which uses a fixed-step RK4 integrator with the same step. The normalization of $\mathbf{q}$ is a separate block to avoid an algebraic loop.

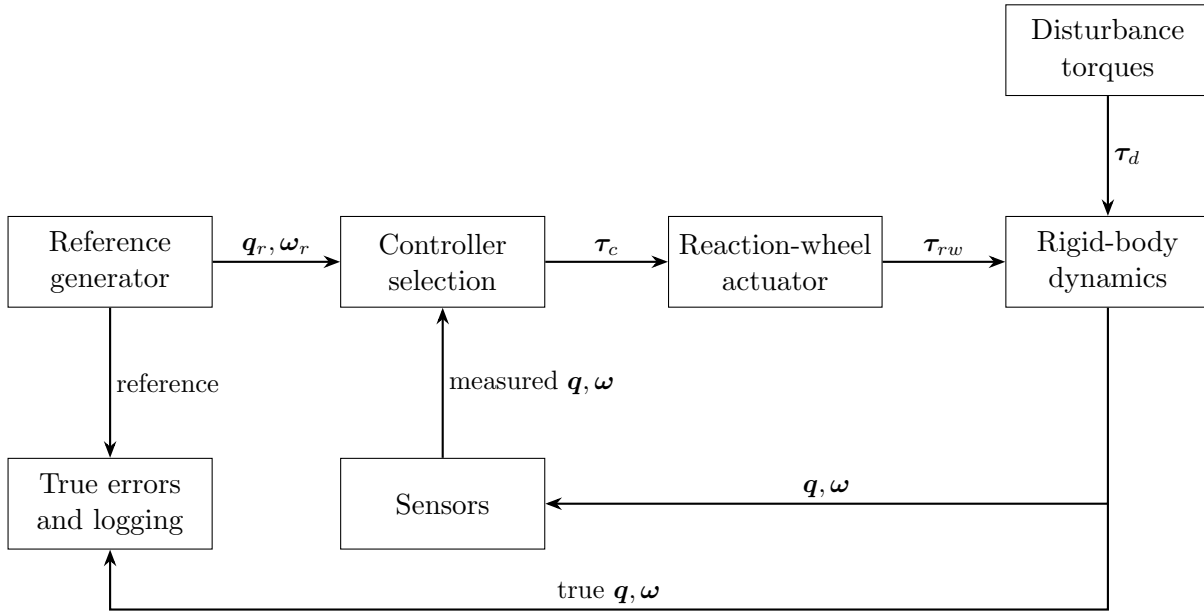


Figure 4.1: Topology of the simulation harness. Every block calls the same library function in the Simulink model and in the reference engine

Table 4.1 summarizes the software layers. The Simulink diagram is wired from a single table, grouped into seven subsystems, and every one of the 43 wires is traced across the subsystem boundaries and verified after the grouping. All MATLAB Function ports have a fixed size and type, which avoids size-propagation errors around the feedback loops.

The nine unit tests cover the quaternion algebra, the reference kinematics, the regressor identity, momentum conservation, the ablation identity between the robust and adaptive controllers, the quaternion norm health, the performance metrics, the configuration input

Table 4.1: Software layers of the MATLAB ADCS Tool

Layer	Content
Simulink model	<code>build_ADCS_model.m</code> generates the harness (<code>ADCS_ComparisonHarness.slx</code> , fixed-step ode4) and the data dictionary <code>ADCS_Params.sldd</code> . 19 signals are logged.
Parameters	Numeric structures <code>SC</code> , <code>RobustGains</code> , <code>AdaptiveGains</code> , <code>BaselineGains</code> and <code>CONTROLLER_SELECT</code> , overridden per run through <code>Simulink.SimulationInput.setVariable</code> . Compare mode runs a batch of three.
Reference engine	<code>simulateADCS.ref.m</code> integrates with RK4 at the same step and calls the same library functions. It is the fallback engine when Simulink is absent and the cross-check when it is present.
GUI	<code>ADCS_ComparisonApp.m</code> , built on <code>matlab.apps.AppBase</code> , with the tabs Scenario & ICs, Controller Gains, Run and Results. Configurations can be saved and loaded.
Tests	Nine unit tests, a scenario smoke test, and an opt-in wiring-repair check that injects a fault into the model build.

and output, and the cross-check between Simulink and the reference engine.

5 Verification

The unit test suite was executed in MATLAB R2026a with Simulink 26.1, and all nine tests pass. The Simulink acceptance gate simulates the nominal scenario for 150 s with every controller on both engines. The maximum attitude-error difference between the engines is 1.7×10^{-11} , 1.2×10^{-11} and $2.7 \times 10^{-11}^\circ$ for the robust, adaptive and PD controllers, against a tolerance of 0.02° , and the RMS wheel torques agree to $10^{-13}\%$. Across all runs the largest quaternion-norm deviation is at the 10^{-10} level. Moreover, torque-free propagation conserves the inertial angular momentum to a relative error of 10^{-8} over 200 s.

The cross-engine scenario check runs all seven scenarios with all three controllers on both engines. Each of the 21 pass or fail verdicts is identical on the two engines, and 20 of the 21 runs pass. Table 5.1 lists the maximum attitude-error difference for each scenario. The residual differences in the saturated slews come from the momentum clamp, which is a limited

integrator in Simulink and a post-step clamp in the reference engine. The difference in the sensor-noise scenario reflects the independent noise realizations of the Simulink random-number blocks and of the MATLAB `randn` function and is not a model difference.

Table 5.1: Simulink versus reference engine in MATLAB R2026a (verdicts listed as robust / adaptive / PD, P = pass, F = fail)

Scenario	Max $ \Delta \text{ attitude error} $ (deg)	Simulink	Reference
Nominal	1.5×10^{-10}	P/P/P	P/P/P
Inertia $\pm 25\%$	2.3×10^{-10}	P/P/P	P/P/P
Large error 150°	2.3×10^{-5}	P/P/P	P/P/P
Disturbance $4\times$	3.7×10^{-11}	P/P/P	P/P/P
Actuator saturation	6.5×10^{-6}	P/P/P	P/P/P
Sensor noise	1.3×10^{-2}	P/P/P	P/P/P
PE scan $\pm 25\%$	5.8×10^{-10}	P/P/F	P/P/F

6 Results

6.1 Scenario Definitions

The seven scenarios are defined in Table 6.1. The initial attitude error is a rotation of the body frame relative to the reference about the stated axis, and every scenario starts with zero rate error and zero wheel momentum. The pass criterion of each scenario is a limit on the steady-state error together with the quaternion norm health ($||\mathbf{q}|| - 1| \leq 10^{-3}$) and finiteness of all signals.

6.2 Performance Metrics

The metrics reported for each run are the settling time t_s , defined as the time after which the attitude error stays within 0.5° , the steady-state error e_{ss} , defined as the mean error over the last 10% of the run, the peak wheel torque τ_{pk} , the control effort $\int \|\boldsymbol{\tau}\|^2 dt$, the peak wheel momentum h_{pk} , the durations of torque saturation and momentum saturation, and

Table 6.1: Scenario definitions and initial conditions

Scenario	Reference mode	Initial error	Duration (s)	Pass limit (deg)	Special settings
Nominal	Docking	5° about $[1, 1, 1]/\sqrt{3}$	400	0.5	Gravity gradient and drag enabled
Inertia $\pm 25\%$	Docking	5° about $[1, 1, 1]/\sqrt{3}$	400	1.0	True inertia perturbed per parameter by up to 25%
Large error 150°	LVLH hold	150° about $[1, 2, 3]/\sqrt{14}$	1500	0.5	Nominal wheel limits
Disturbance $4\times$	Docking	5° about $[1, 1, 1]/\sqrt{3}$	600	2.0	All four disturbances enabled and scaled by 4
Actuator saturation	LVLH hold	120° about $[1, 2, 3]/\sqrt{14}$	2000	1.0	$\tau_{\max} = 0.075$ N m, $h_{\max} = 7.5$ N m s
Sensor noise	Docking	5° about $[1, 1, 1]/\sqrt{3}$	400	1.0	Noisy star tracker and gyro
PE scan $\pm 25\%$	Attitude scan	5° about $[1, 1, 1]/\sqrt{3}$	1600	0.5	Scan amplitudes 3° , frequencies 0.015, 0.020, 0.025 rad/s; inertia $\pm 25\%$; $\Gamma = 5 \times 10^9$

the final inertia-estimate error $\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|/\|\boldsymbol{\theta}\|$ for the adaptive controller. Table 6.2 reports the accuracy and settling results, and Table 6.3 reports the actuator demand. Both tables are produced by the Simulink engine.

6.3 Scenario Responses

Figure 6.1 compares the steady-state error of the three controllers across the scenarios on a logarithmic scale. Figure 6.2 shows the nominal docking approach, in which all controllers briefly saturate the wheel torque while the wheel momentum stays far below $h_{\max}$. Figure 6.3 shows the 150° slew, in which the wheel momentum saturates and the maneuver becomes rate-limited to about $h_{\max}/J \approx 0.005$ rad/s, so that settling takes about 550 to 600 s for every controller. Figure 6.4 shows the adaptation windup during the same slew, and Figure 6.5

Table 6.2: Accuracy and settling results per scenario and controller (Simulink engine, MATLAB R2026a)

Scenario	Controller	t_s (s)	e_{ss} (deg)	$\ \tilde{\theta}\ $ (%)	Pass
Nominal	Robust SMC	119	0.040	-	P
	Adaptive SMC	119	0.041	0.5	P
	PD	110	0.090	-	P
Inertia $\pm 25\%$	Robust SMC	118	0.030	-	P
	Adaptive SMC	118	0.031	13.8	P
	PD	110	0.073	-	P
Large error 150°	Robust SMC	591	0.046	-	P
	Adaptive SMC	589	0.046	1.4	P
	PD	536	0.085	-	P
Disturbance $4\times$	Robust SMC	119	0.163	-	P
	Adaptive SMC	118	0.164	0.5	P
	PD	111	0.264	-	P
Actuator saturation	Robust SMC	876	0.046	-	P
	Adaptive SMC	876	0.046	0.3	P
	PD	833	0.085	-	P
Sensor noise	Robust SMC	119	0.050	-	P
	Adaptive SMC	119	0.051	0.5	P
	PD	110	0.101	-	P
PE scan $\pm 25\%$	Robust SMC	132	0.150	-	P
	Adaptive SMC	144	0.019	1.8	P
	PD	-	1.461	-	F

shows the persistently exciting scan.

7 Discussion

Robust SMC versus the benchmark

Both sliding-mode variants reduce the steady-state error relative to the PD benchmark in every scenario, for example 0.040° versus 0.090° in the nominal case and 0.163° versus 0.264° in the disturbance stress case, as listed in Table 6.2. This follows from the model-based feedforward and the sliding-surface structure, whereas the proportional action of the PD law

Table 6.3: Actuator demand per scenario and controller (Simulink engine, MATLAB R2026a)

Scenario	Controller	τ_{pk} (N m)	$\int \ \tau\ ^2 dt$ (N ² m ² s)	h_{pk} (N m s)	τ -sat (s)	h -sat (s)
Nominal	Robust SMC	0.150	0.43	2.0	4	0
	Adaptive SMC	0.150	0.43	2.0	4	0
	PD	0.141	0.32	1.9	0	0
Inertia $\pm 25\%$	Robust SMC	0.150	0.48	2.2	4	0
	Adaptive SMC	0.150	0.48	2.2	4	0
	PD	0.141	0.36	2.0	0	0
Large error 150°	Robust SMC	0.150	12.47	15.0	492	348
	Adaptive SMC	0.150	12.48	15.0	492	348
	PD	0.150	10.51	15.0	451	319
Disturbance $4\times$	Robust SMC	0.150	0.44	2.6	4	0
	Adaptive SMC	0.150	0.44	2.6	4	0
	PD	0.141	0.33	2.6	0	0
Actuator saturation	Robust SMC	0.075	2.95	7.5	785	657
	Adaptive SMC	0.075	2.95	7.5	785	657
	PD	0.075	2.63	7.5	747	631
Sensor noise	Robust SMC	0.150	0.44	2.0	4	0
	Adaptive SMC	0.150	0.44	2.0	4	0
	PD	0.142	0.33	1.9	0	0
PE scan $\pm 25\%$	Robust SMC	0.150	13.97	9.6	6	0
	Adaptive SMC	0.150	13.72	9.6	6	0
	PD	0.144	18.68	10.4	0	0

alone leaves a bias against the gravity-gradient and drag torques. The PD benchmark settles slightly earlier because of its larger proportional authority.

What adaptation provides

In the $\pm 25\%$ inertia docking case, the adaptive and robust controllers are practically identical (0.031° versus 0.030°) and the estimate $\hat{\theta}$ hardly moves. The LVLH-referenced reference is nearly stationary, so the regressor is not persistently exciting [19], and the high-gain sliding feedback already absorbs the inertia error. With the PE scan of Figure 6.5 the behavior changes: the steady-state errors are 0.019° for the adaptive controller, 0.150° for

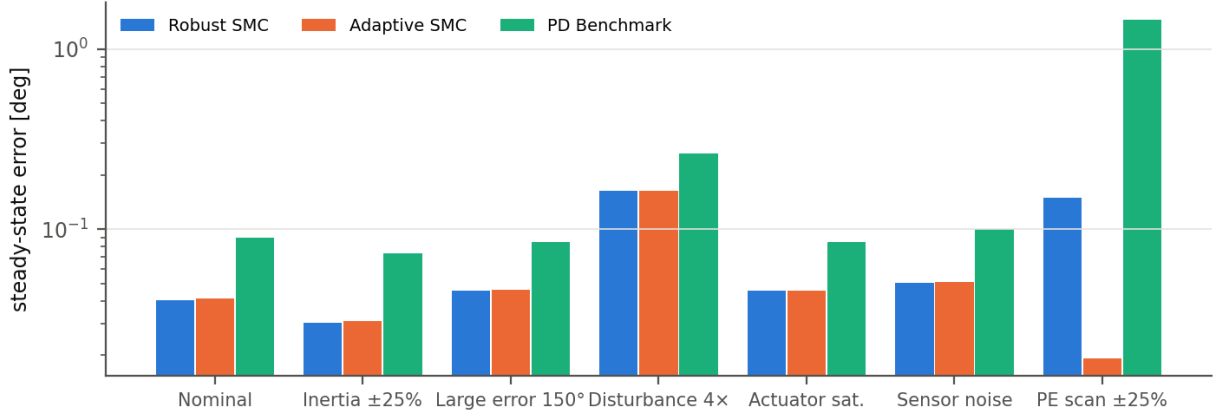


Figure 6.1: Steady-state attitude error by scenario (logarithmic scale)

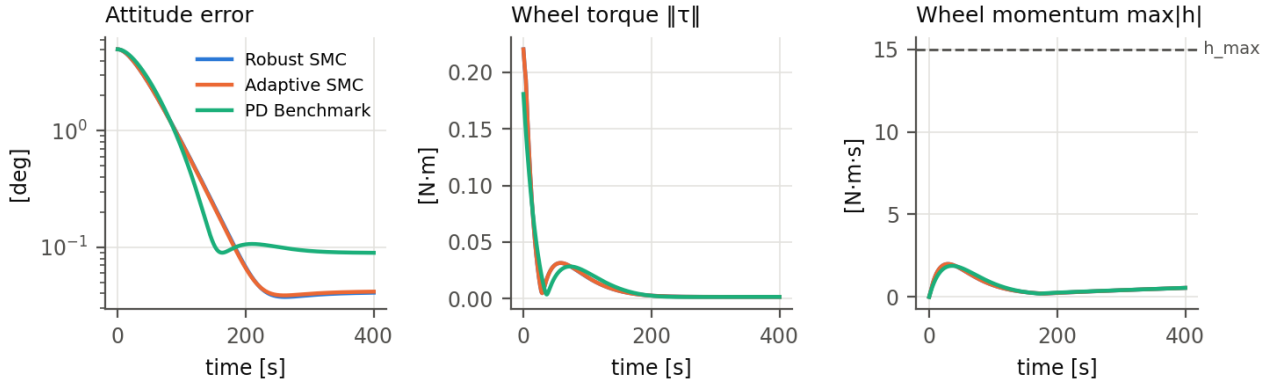


Figure 6.2: Nominal docking approach with a 5° initial error

the robust controller and 1.461° for the PD benchmark, and the principal moments converge to within a few percent of the true values. The adaptive controller is therefore about 8 times more accurate than the robust controller and about 75 times more accurate than the PD benchmark in this scenario.

Adaptation gain versus excitation

A gain sweep with $\Gamma = 10^9$ and 5×10^9 in every non-PE scenario shows $\hat{\theta}$ drifting 15 to 49% from the truth even when the true inertia equals the nominal value. The estimator absorbs unmodelled disturbance torques in the absence of excitation, although the settling becomes

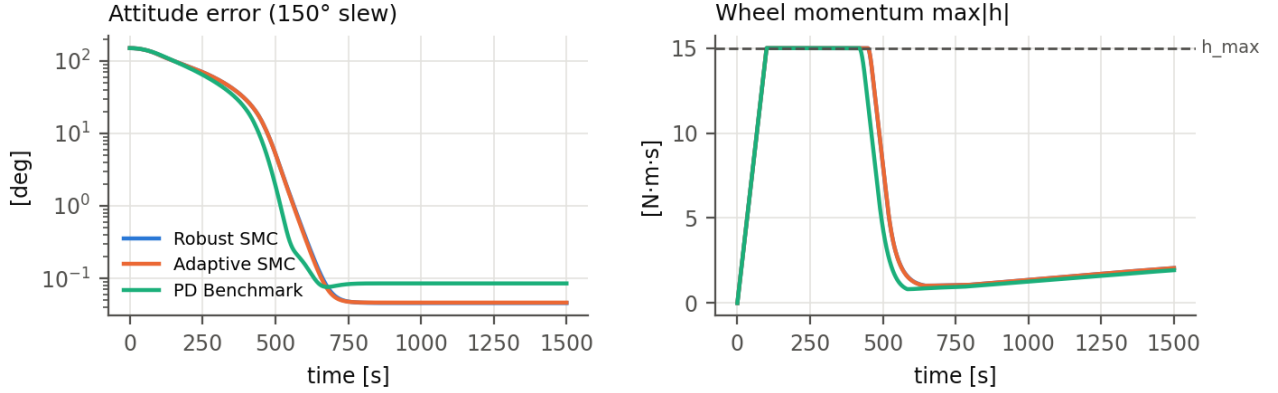


Figure 6.3: 150° slew with wheel momentum saturation

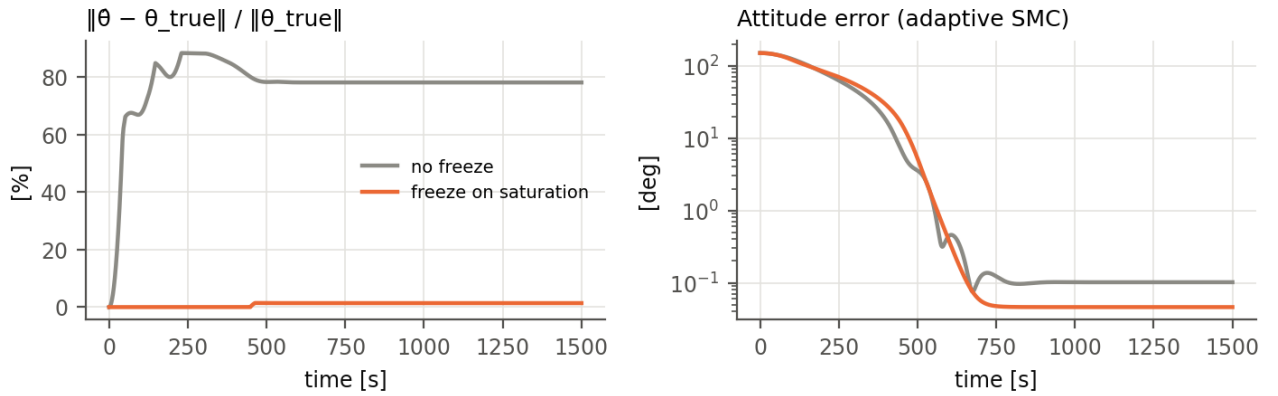


Figure 6.4: Adaptation windup during the 150° slew, without and with the freeze-on-saturation logic

faster (81 s versus 119 s). The shipped default is therefore a conservative $\Gamma = 2 \times 10^7$, and the PE scenario raises it explicitly to 5×10^9 . An excitation-aware adaptation, using PE monitoring, composite or concurrent learning, or dead zones, is the principled remedy for this trade-off.

Actuator sizing

With $h_{\max} = 15$ N·m·s and inertias between about 1200 and 3100 kg·m², large slews are momentum-limited regardless of the controller, as shown by the saturation durations in Table 6.3. In practice, the actuator therefore dominates the comparison in these scenarios. The

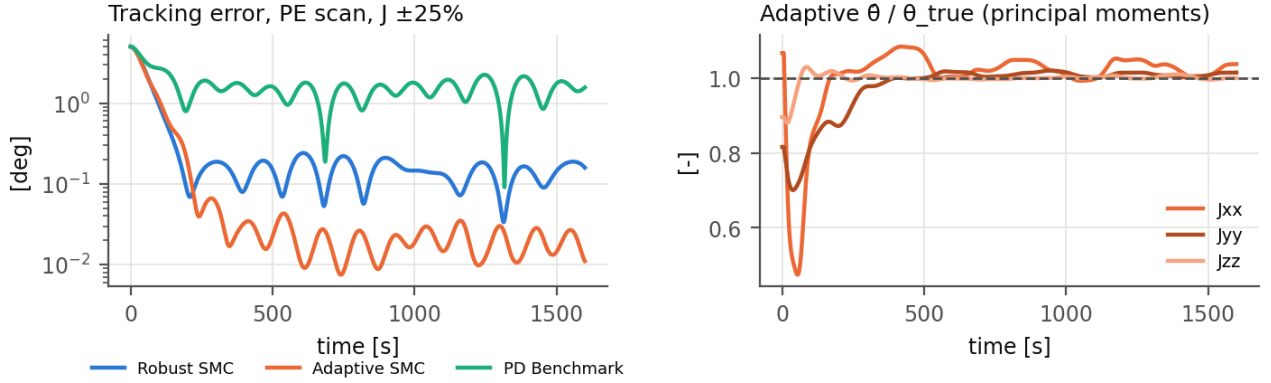


Figure 6.5: Persistently exciting attitude scan with a $\pm 25\%$ inertia error

performance differences between the controllers appear in the accuracy and the robustness, not in the slew time.

8 Conclusions and Future Work

This work compared a robust sliding-mode controller, an adaptive sliding-mode controller with inertia estimation, and a quaternion PD benchmark on a LEO chaser tracking a line-of-sight reference to the ISS. A programmatically generated Simulink model and an independent reference engine reproduce each other to about 10^{-11}° , and 20 of 21 scenario runs meet their pass criteria. The only failure is the PD benchmark in the persistently exciting scan. The sliding-mode variants halve the steady-state error of the benchmark in the docking scenarios, and the adaptive controller improves the tracking error about 8-fold over the robust controller only when the reference excites the inertia regressor.

Future work includes terminal and prescribed-time sliding-mode laws [20, 21], hybrid hysteretic quaternion feedback to remove the chattering of the shortest-path switch near 180° [22], input-constrained and excitation-aware adaptation [17, 18], a four-wheel pyramid configuration with wheel faults, coupled 6DOF relative motion [9], estimation with a multiplicative extended Kalman filter [23], and Monte Carlo robustness envelopes.

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